deviations from the mean, the value of the Gaussian curve has dropped to about 1/19, 1/7563, and 1/166,666,666, respectively. This is why normally distributed signals, such as illustrated in Fig. 2-6c, *appear* to have an approximate peak-to-peak value. In principle, signals of this type can experience excursions of unlimited amplitude. In practice, the sharp drop of the Gaussian pdf dictates that these extremes almost never occur. This results in the waveform having a relatively bounded appearance with an apparent peak-to-peak amplitude of about 6-8σ.

As previously shown, the integral of the pdf is used to find the probability that a signal will be within a certain range of values. This makes the integral of the pdf important enough that it is given its own name, the **cumulative distribution function (cdf)**. An especially obnoxious problem with the Gaussian is that it cannot be integrated using elementary methods. To get around this, the integral of the Gaussian can be calculated by *numerical integration*. This involves sampling the continuous Gaussian curve very finely, say, a few million points between -10σ and +10σ. The samples in this discrete signal are then *added* to simulate *integration*. The discrete curve resulting from this simulated integration is then stored in a table for use in calculating probabilities.

The cdf of the normal distribution is shown in Fig. 2-9, with its numeric values listed in Table 2-5. Since this curve is used so frequently in probability, it is given its own symbol: $\Phi(x)$ (upper case Greek *phi*). For example, $\Phi(-2)$ has a value of 0.0228. This indicates that there is a 2.28% probability that the value of the signal will be between -∞ and two standard deviations below the mean, at any randomly chosen time. Likewise, the value: $\Phi(1) = 0.8413$, means there is an 84.13% chance that the value of the signal, at a randomly selected instant, will be between -∞ and one standard deviation above the mean. To calculate the probability that the signal will be will be *between* two values, it is necessary to subtract the appropriate numbers found in the $\Phi(x)$ table. For example, the probability that the value of the signal, at some randomly chosen time, will be between two standard deviations below the mean and one standard deviation above the mean, is given by: $\Phi(1) - \Phi(-2) = 0.8185$, or 81.85%

Using this method, samples taken from a normally distributed signal will be within ±1σ of the mean about 68% of the time. They will be within ±2σ about 95% of the time, and within ±3σ about 99.75% of the time. The probability of the signal being more than 10 standard deviations from the mean is so minuscule, it would be expected to occur for only a few *microseconds* since the beginning of the universe, about 10 billion years!

Equation 2-8 can also be used to express the probability mass function of normally distributed *discrete* signals. In this case, x is restricted to be one of the quantized levels that the signal can take on, such as one of the 4096 binary values exiting a 12 bit analog-to-digital converter. Ignore the $1/\sqrt{2\pi}\sigma$ term, it is only used to make the total area under the pdf curve equal to *one*. Instead, you must include whatever term is needed to make the sum of all the values in the pmf equal to *one*. In most cases, this is done by